

A Hierarchical Framework-Based Coordinated Optimization of Building HVAC Systems and EVs

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Abstract—The demands of electric vehicles (EVs) and building heating, ventilation, and air conditioning (HVAC) systems have considerable flexibility. Their flexibility is influenced by occupants' behavior resulting in huge complementary and dispatchable capability. Therefore, the coordination of EVs and HVAC systems holds significant potential for optimizing the building demand profiles and energy cost under time-of-use (TOU) tariffs. However, solving the coordinated problem in practice still faces the challenges in computational complexity and global information requirement due to the spatio-temporal coupling constraints. In this paper, a mixed-integer linear programming model is developed to formulate the multi-building energy system with EVs and the impact of occupants' behavior on the demand and flexibility of the system. The problem is converted into a three-level structure using Lagrangian relaxation framework. A dynamic programming-based Lagrangian relaxation (DPLR) algorithm is developed to independently solve all sub-problems of the three-level structure in a decomposition and coordination way while avoiding the iterative computation between the middle and lower level. The numerical results show the developed method can obtain a near-optimal solution in an efficient way without perceivable degradation in accuracy, which is 4% worse but at least three times faster, compared to the existing centralized algorithm.

Note to Practitioners—The escalating demand for EVs and HVAC systems results in increased energy costs and challenges to existing power systems, such as frequency deviations and higher peak loads. Therefore, this paper focuses on the coordinated optimization of the EV charging and building HVAC system operation, while considering rooftop photovoltaic generation supply within the system. Optimal coordination of the above system can effectively reduce energy costs under TOU tariffs and enhance the ability to utilize renewable energy sources. However, solving the coordinated optimization problem still faces difficulties since the computational complexity will exponentially grow with increasing problem scale due to the spatio-temporal coupling between the demand of EVs and HVAC systems.

Therefore, a DPLR algorithm is developed. There is a central coordinator that collects the demand information of EVs and HVAC systems and broadcasts the coordination information calculated according to the demand information. Every single EV and HVAC system can make decisions based on the coordination information independently. The DPLR algorithm decouples EVs and HVAC systems and avoids the curse of dimensionality. It has great improvement on computational efficiency and global information requirement reduction. The computational efficiency and effectiveness of the developed method is verified through multi-scale case studies. The numerical results show that compared with independent optimization of EVs and HVAC systems, coordinated optimization can reduce over 44% of the energy costs.

Index Terms—Demand side management, coordinated optimization, electric vehicle (EV) and building HVAC system, Lagrangian relaxation (LR), dynamic programming.

NOTATION

A. Indices and Sets

A^R	Set of the actions of building HVAC system.
A_k	Set of the actions at time stage k .
I	Set of the index for EVs.
J	Set of the index for rooms.
J_n^B	Set of the index for rooms in building n .
K	Set of the index for time stages.
N	Set of the index for buildings.
S_k	Set of the system states at time stage k .
i	Index of EVs, $i \in \{1, 2, \dots, I\}$.
j	Index of rooms, $j \in \{1, 2, \dots, J\}$.
k	Index of time stages, $k \in \{1, 2, \dots, K\}$.
n	Index of buildings, $n \in \{1, 2, \dots, N\}$.

B. Parameters

Δk	Time period per stage (h).
η_j	The coefficient of performance of the HVAC system in room j .
$\underline{S}$	Minimum state of charge of EVs.
C_j	Equivalent heat capacity of room j (kJ/°C).
$c_k^p(c_k^s)$	Electricity price purchased from (sold to) power grid at time stage k (CNY/kWh).
$E_i^{V, cap}$	Battery capacity of EV i (kWh).
$f_{k,i}(\Delta k)$	The energy consumption of EV i for driving a time stage at time stage k .
I	Total number of EVs.

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J	Total number of rooms.
K	Total number of scheduling time stages.
N	Total number of buildings.
$O_{k,j}$	Occupied state of room j at time stage k .
P_i^V	Rated charging and discharging power of EV i (kW).
P_n^{cap}	Power capacity between the power grid and building n (kW).
$P_n^{EV, cap}$	Power capacity of EV charging facility in building n (kW).
$P_{k,n}^e$	Power load of other electrical equipments in building n at time stage k (kW).
$P_j^{HVAC, cap}$	Rated capacity of HVAC system in room j (kW).
$P_{k,n}^{PV}$	Power supplied by roof-top photovoltaic generation in building n at time stage k (kW).
R_j^e	Equivalent thermal resistance of room j ($^{\circ}\text{C}/\text{kW}$).
$T^-(T^+)$	Lower (Upper) indoor temperature bound for human thermal comfort requirement ($^{\circ}\text{C}$).
T_k^o	The outdoor air temperature at time stage k ($^{\circ}\text{C}$).

C. Variables

$a_{k,j}^R$	HVAC system action of room j at time stage k (kW).
$a_{k,i}^{V,c} (a_{k,i}^{V,d})$	Charging (discharging) action of EV i at time stage k , $a_{k,i}^{V,c} \in \{0, 1\}$, $a_{k,i}^{V,d} \in \{0, 1\}$.
$P_{k,n}^{EV}$	Power exchange between building n and EV charging facility at time stage k (kW).
$P_{k,n}^P (P_{k,n}^S)$	Power of building n purchased from (sold to) power grid at time stage k (kW).
$S_{k,i}$	The state of charge of EV i at time stage k , $S_{k,i} \in [0, 1]$.
$T_{k,j}^R$	The indoor temperature of room j at time stage k ($^{\circ}\text{C}$).

I. INTRODUCTION

AFTER the remarkable development of electric vehicle (EV) over the past decade, the total number of EVs have been raising to 26 million in 2022 [1]. However, the increasing number of EVs has resulted in a significant increase in electricity consumption and brought challenges to power grids [2], such as frequency deviation and higher peak loads [3], [4]. On the other hand, building energy systems (BES) account for over 40% of energy consumption in the U.S., among them, heating, ventilation, and air conditioning (HVAC) systems consume 45% of the electricity energy [5]. Due to room indoor temperature dynamics and EVs' required charging time being much shorter than their parking duration, the coordination of EVs and HVAC systems provides huge potential in enhancing the flexibility of BES for demand side management [6].

To coordinately optimize the operation of EVs and BES, various studies have been made to minimize their total cost [7], [8], improve the utilization efficiency of distributed renewable energy sources [9], reduce peak load and reshape demand profiles [10]. EVs and buildings typically share the

same infrastructure, Contreras-Ocana et al. [11] proposed a coordinated optimization problem of EVs and buildings considering the demand limits, peak demand charges and itemized billing tariffs to optimize their demand flexibility. Wang et al. [12] developed a three-level scheduling model for aggregated HVAC systems and EVs under the locational marginal price in the competitive market to make benefit for both distribution systems and aggregators.

The coordinated problem of EVs and HVAC systems is usually large-scale. Many efforts have been made to tackle the curse of dimensionality. E.g., Hurtado et al. [13] developed a multi-agent based decentralized approach to optimize the EVs charging strategy and reshape the demand profile of BES. Yang et al. [14] developed a decentralized EV-based charging algorithm under the Lagrangian relaxation (LR) framework to determine multiple EVs charging based on the building's electrical and thermal demand. In order to improve the computational efficiency by reducing the state space, the event-based approaches are also developed to solve the optimization problem of EVs and building HVAC systems, since the state space aggregation can be realized by defining several events [15], [16], [17]. Furthermore, other decentralized algorithms, such as Dantzig-Wolfe decomposition [18], Bender's decomposition [19], and the alternating direction method of multipliers algorithm (ADMM) [20] are also developed to solve the coordinated problem. Although many innovative results in the coordinated optimization of EVs and building HVAC systems are reported in the literature, the computational efficiency and convergence of existing methods are still hard to be guaranteed due to the spatio-temporal coupling constraints caused by occupant's thermal comfort requirement and mobility.

Thus, there are still many challenges in the coordination of EVs and building HVAC systems. First, spatio-temporal coupling. Multiple buildings are coupled in space due to the mobility of EVs. In addition, the operation of EVs and HVAC are also coupled in time due to the thermal comfort and next trip requirement of occupants. So, the above spatio-temporal coupling between multiple buildings and EVs needs to be well formulated to maximally enhance the flexibility provided by EVs and HVAC systems. Second, multiple stakeholders. Multiple stakeholders of EVs and buildings are involved in the energy conversion and interaction process of the coordination of EVs and HVAC systems. The centralized algorithm is thus limited since the global information is costly to collect in practice due to occupants' privacy concerns and high communication and computation costs. Third, the curse of dimensionality. The state space and policy space of the coordinated problem will exponentially grow with the increasing scale of EVs and buildings. The coordinated problem is believed to be NP-hard [14], [15]. In addition, due to the spatio-temporal coupling, the existing decentralized method cannot be directly applied since their optimality and convergence may not be guaranteed. Therefore, an effective and efficient decentralized optimization method needs to be developed based on the structure of the coordinated problem for practical engineering.

In order to overcome the challenges mentioned above, we make the following contributions in this paper. First,

the coordinated optimization problem of multiple EVs and building HVAC systems is formulated as a centralized mixed-integer linear programming (MILP) problem. The spatio-temporal coupling and interaction between EVs and HVAC systems are captured based on power balance with occupants' thermal comfort requirement of each building and mobility behavior of each EV. In this way, the coupling and interaction between HVAC systems and each EV are formulated, thus, the system flexibility can be enhanced with the coordination of buildings and all the EVs. Second, an LR-based decentralized optimization framework is developed to convert the original centralized problem into a problem with three-level structure. The EVs and HVAC systems are decoupled by introducing Lagrangian multipliers and relaxing the system-wide constraints. The problem can be solved with inner and outer iteration in a decomposition and coordination way that reduce the global information requirement. Third, a dynamic programming-based LR (DPLR) algorithm is developed to efficiently solve the problem of the three-level structure problem. Based on the DPLR algorithm, the optimal analytic solution of EV charging sub-problems in the lower level are derived by dynamic programming. The dual problems of EV charging facility operation sub-problem in the middle level are transformed from "max-min" structure problems to "max" optimization problems of the Lagrangian multipliers, thus avoiding the inner iteration between the middle and lower level. In this way, the sub-problems of each HVAC system operation and the sub-problems of each EV charging can be independently solved. Numerical results show the coordinated optimization can get better utilization of the complementary flexibility of EVs and HVAC systems demand and save over 44% of the energy cost. The performance of the DPLR algorithm developed in this paper has great improvement that at least three times faster than the centralized algorithm with less than 4% worse accuracy. It also can save 52% computational time with better accuracy compared with the traditional LR algorithm.

The remainder of this paper is organized as follows. In Section II, the coordinated problem is formulated as a centralized MILP problem. In Section III, the decentralized problem of a three-level structure is developed based on the LR framework and the DPLR algorithm is developed. In Section IV, the case studies and numerical results are presented to validate the performance of the DPLR algorithm. The discussion of the performance of developed DPLR algorithm is given in Section V. A brief conclusion is given in Section VI.

II. PROBLEM FORMULATION

This paper focuses on the coordination of multi-building energy system with EVs as shown in Fig. 1. There may be various kinds of buildings, e.g., residential buildings, office buildings and commercial buildings in the system. Within each building, there are amount of rooms with HVAC system, parking lots with charging facility, roof-top photovoltaic generation devices and other electrical equipments. There are bidirectional energy interactions between EVs and each building, including EV charging from buildings and feeding energy into buildings through charging facility in buildings.

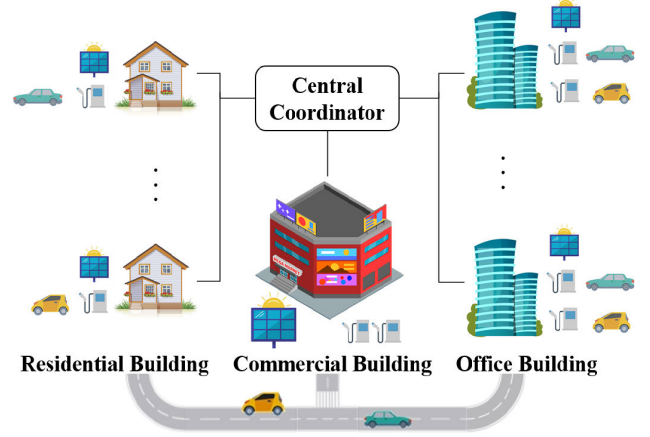


Fig. 1. Multi-building energy system with EVs.

As similar to [21] and [22], we assume that there is a central coordinator among buildings in the system (e.g., the aggregator of buildings or the scheduling center of the city or town). It is connected with all buildings to collect global information (such as the trip plans and initial state of charge of EVs, and the power supplied by roof-top photovoltaic generation) and broadcasts centralized coordination information to help all buildings make decision for their operation of HVAC systems and EVs charging, while satisfying the occupant thermal comfort requirement and the trip energy requirement of occupant mobility in daily life.

In this section, the coordinated optimization problem of the multi-building energy system with EVs is formulated as a discrete-time-based MILP problem. The objective is to minimize the energy cost of the multi-building energy system under TOU tariff mechanism while satisfying the occupant's thermal comfort requirement and mobility requirement. In the following, the objective function is given in sub-section III-A. The operation constraints including the coupling between occupant thermal comfort and mobility requirement are developed in sub-section III-B. The centralized coordinated optimization problem is described in sub-sections III.C.

A. Objective Function

The objective of the coordinated problem is to minimize the energy cost of the system while satisfying occupants' requirement. The EVs can only charging and discharging in the buildings. So, the energy cost of EVs is considered in the building's energy cost. Thus, the objective function during the scheduling horizon is defined as

$$\min C = \sum_{k=1}^K \sum_{n=1}^N (c_k^p P_{k,n}^p - c_k^s P_{k,n}^s) \Delta k. \quad (1)$$

B. System Constraints

To guarantee the power balance of building n , we have

$$P_{k,n}^p - P_{k,n}^s + P_{k,n}^{PV} = P_{k,n}^e + P_{k,n}^{EV} + \sum_{j \in J_n^B} a_{k,j}^R. \quad (2)$$

The building HVAC systems and EV charging facility in building n are coupled by this constraint.

Intuitively, the power purchased and sold by buildings should be constrained by the power capacity of the transmission line. Besides, buildings cannot purchase and sell electricity to the grid at the same time for safety reason, i.e.,

$$0 \leq P_{k,n}^p \leq z_{k,n}^p P_n^{cap}, \quad (3)$$

$$0 \leq P_{k,n}^s \leq z_{k,n}^s P_n^{cap}, \quad (4)$$

$$z_{k,n}^p + z_{k,n}^s \leq 1, \quad (5)$$

where $z_{k,n}^p \in \{0, 1\}$ and $z_{k,n}^s \in \{0, 1\}$ are binary variables that $z_{k,n}^p = 1$ ($z_{k,n}^s = 1$) indicating building n purchases (sells) power from (to) power grid at time stage k . There is no energy exchange between building n and power grid when $z_{k,n}^p = 0$ and $z_{k,n}^s = 0$.

The total power exchanged between building n and EV charging facility at time stage k consists of the charging and discharging power of the EVs parking in building n . It should be constrained by charging facility capacity, i.e.,

$$P_{k,n}^{EV} = \sum_{i=1}^I L_{k,i,n} (a_{k,i}^{V,c} - a_{k,i}^{V,d}) P_i^V, \quad (6)$$

$$-P_n^{EV, cap} \leq P_{k,n}^{EV} \leq P_n^{EV, cap}, \quad (7)$$

where $L_{k,i,n}$ is a binary variable indicating the location of EV i at time stage k is building n or not and $L_{k,i,n} \in \{0, 1\}$. $L_{k,i,n} = 1$ ($L_{k,i,n} = 0$) indicates EV i is (is not) parking in building n at time stage k . $\sum_{n=1}^N L_{k,i,n} = 0$ indicates EV i is on the road at time stage k . Actually, $L_{k,i,n}$ and P_i^V are parameters. Therefore, the constraint (6) is linear with respect to the actions $a_{k,i}^{V,c}$ and $a_{k,i}^{V,d}$. Equation (6) implies that $P_{k,n}^{EV}$ only take into account the EVs parking in building n at time stage k . Similar to [22], the charging and discharging rate of EVs is assumed to be a constant. Therefore, the charging and discharging action of EVs are defined as binary variables that $a_{k,i}^{V,c} = 1$ ($a_{k,i}^{V,d} = 1$) means EV i is charged (discharged) at time stage k . Otherwise, $a_{k,i}^{V,c} = 0$ and $a_{k,i}^{V,d} = 0$ indicates the EV i is idle at time stage k . Based on (6), EVs are coupled together because they share the same charging facility in the building. Moreover, the buildings are also coupled due to EVs charge and discharge in different buildings that caused by the mobility of EVs. Equation (7) implies the power exchanged between building and EV charging facility cannot exceed the charging facility capacity.

The dynamics of EV's state of charge (SoC) is given by:

$$S_{k+1,i} = S_{k,i} + \frac{(a_{k,i}^{V,c} - a_{k,i}^{V,d}) P_i^V \Delta k}{E_i^{V, cap}} - (1 - z_{k,i}^{EV}) \times \frac{f_{k,i}(\Delta k)}{E_i^{V, cap}}, \quad (8)$$

where the parking state $z_{k,i}^{EV} \in \{0, 1\}$ denotes the EV i is parking at buildings ($z_{k,i}^{EV} = 1$) or trip on the road ($z_{k,i}^{EV} = 0$) at time stage k . The energy consumption of EV i for driving a time stage at time stage k is denoted as $f_{k,i}(\Delta k)$. Note that the second term of the right-hand side represents the energy change caused by charging (discharging) action when the EV parks in a building and the last term is the energy consumed by the trip when EV is on the road.

The relationship between the location and parking state of EV i at time stage k can be denoted as:

$$z_{k,i}^{EV} = \sum_{n=1}^N L_{k,i,n}. \quad (9)$$

Considering the constraints of EV battery capacity and charging (discharging) location, we have:

$$\underline{S} \leq S_{k,i} \leq 1, \quad (10)$$

$$a_{k,i}^{V,c} + a_{k,i}^{V,d} \leq z_{k,i}^{EV}. \quad (11)$$

Equation (10) implies the SoC of EVs cannot below the lower limit, for the purpose of battery life-time concern, and cannot exceed the battery capacity. Equation (11) implies that the EVs cannot be charged or discharged ($a_{k,i}^{V,c} = 0$, $a_{k,i}^{V,d} = 0$) when they trip on the road ($z_{k,i}^{EV} = 0$) and cannot be charged and discharged simultaneously.

Furthermore, in order to formulate the dynamics of the indoor temperature of the room, a resistance-capacitance circuit model [23] is adopted, i.e.,

$$T_{k+1,j}^R = T_k^o - \eta_j a_{k,j}^R R_j^e + (T_{k,j}^R - T_k^o + \eta_j a_{k,j}^R R_j^e) e^{-\frac{\Delta k}{R_j^e C_j}}. \quad (12)$$

The HVAC system is assumed to be a multi-speed system so that the $a_{k,j}^R$ is continuous and is constrained by rated capacity of building HVAC system [24]:

$$0 \leq a_{k,j}^R \leq P_j^{HVAC, cap}. \quad (13)$$

As mentioned before, the HVAC system should be operated to satisfy human thermal comfort requirement when the room is occupied, i.e.,

$$-M(1 - O_{k,j}) + O_{k,j} T^- \leq T_{k,j}^R \leq O_{k,j} T^+ + M(1 - O_{k,j}), \quad (14)$$

where M is a big positive number to constrain the indoor temperature under unoccupied stages and $O_{k,j}$ denotes the occupied state of the room j at time stage k i.e.,

$$O_{k,j} = \gamma_{k,i,j} I(L_{k,i,n} = 1) + \nu_{k,i,j} (1 - I(L_{k,i,n} = 1)), \quad (15)$$

where $\gamma_{k,i,j}$ and $\nu_{k,i,j}$ are random binary variables that denote the occupied state of room j in building n ($j \in J_n^B$) which has the same occupant with EV i . $\gamma_{k,i,j} = 1$ ($\gamma_{k,i,j} = 0$) indicates that the room j is occupied (not occupied) when the EV i is parking in building n at time stage k . Similarly, $\nu_{k,i,j} = 1$ ($\nu_{k,i,j} = 0$) indicates that the room j is occupied (not occupied) when the EV i is not parking in building n at time stage k . The $\gamma_{k,i,j}$ and $\nu_{k,i,j}$ can be obtained by the history data of EVs location and room occupied state. This constraint couples the demand of EVs and HVAC systems.

C. Coordinated Optimization Problem

A daily coordinated optimization problem for N buildings, J rooms and I EVs is formulated, which is uniformly divided into K time stages. The system states at stage k including the SoC of EVs and the indoor temperature of rooms are defined as $S_k = [S_{k,i}, T_{k,j}^R], \forall i \in I, \forall j \in J$. The system decisions

including the power purchased and sold by buildings, the action of building HVAC system operation and the charging and discharging action of EVs at stage k are denoted as $A_k = [P_{k,n}^p, P_{k,n}^s, a_{k,j}^R, a_{k,i}^{V,c}, a_{k,i}^{V,d}]$, $\forall n \in N, \forall j \in J, \forall i \in I$.

Overall, the coordinated problem of EVs and HVAC systems can be written as the centralized optimization problem **P1**, i.e.,

$$\begin{aligned} \mathbf{P1}: \min_A C, \text{ with } C &= \sum_{n=1}^N \sum_{k=1}^K (c_k^p P_{k,n}^p - c_k^s P_{k,n}^s) \Delta k \\ \text{s.t. } &(2) - (15). \end{aligned} \quad (16)$$

The optimal operation strategy of EVs and building HVAC systems to minimize the energy cost of the multi-building energy system with EVs can be obtained by solving the problem **P1**. However, due to the spatio-temporal coupling constraints in (2) and (6), the state space at time stage k is $\mathcal{S}_k \subset \mathbb{R}^I \times \mathbb{R}^J$. The decision space of EVs is $2^{I \times K} \times 2^{I \times K}$ and the decision space of HVAC systems is $A^R = (A_1^R, A_2^R, \dots, A_K^R) \subset \mathbb{R}^{J \times K}$. The decision space in the centralized problem is the decision space of EVs times the decision space of HVAC systems that will exponentially grow with the increasing scale of the problem. Solving such optimization problem faces the computational complexity challenge. Furthermore, the central coordinator has to collect all the global information for the purpose of solving the centralized problem. In practice, collecting global information may face difficulties in high transmission requirement and occupants' privacy concerns. Therefore, A DPLR algorithm is developed in Section III to let EVs and HVAC systems make their decision in a decomposition and coordination way.

III. HIERARCHICAL FRAMEWORK-BASED OPTIMIZATION ALGORITHM

As mentioned before, solving the centralized coordinated problem **P1** directly faces the challenges in computational complexity and global information requirement in practical engineering. In order to tackle the challenges, **P1** is equivalently decomposed into a hierarchical framework with three levels based on Lagrangian relaxation in sub-section III-A. A DPLR algorithm is developed to solve the hierarchical framework-based problem in subsection III-B.

A. Lagrangian Relaxation Framework

As shown in (1), the objective function has decomposable structure for different buildings but EVs and the HVAC systems in same building are coupled with (2). The Lagrangian function is defined by introducing the Lagrangian multipliers λ to relax the coupled constraint (2), i.e.,

$$\begin{aligned} L &= \sum_{k=1}^K \sum_{n=1}^N (c_k^p P_{k,n}^p - c_k^s P_{k,n}^s) \Delta k + \sum_{k=1}^K \sum_{n=1}^N \lambda_{k,n} (P_{k,n}^p \\ &\quad - P_{k,n}^s + P_{k,n}^{PV} - P_{k,n}^e - P_{k,n}^{EV} - \sum_{j \in J_n^B} a_{k,j}^R), \end{aligned} \quad (17)$$

where $\lambda = \{\lambda_{k,n}\}$, $\forall n \in N, \forall k \in K$ is the set of Lagrangian multipliers.

Therefore, using the duality theorem [25], the coordinated problem **P1** is decomposed to the dual problems of **P1** and the sub-problems of building operation, EV charging facility operation and HVAC system operation. The sub-problems of building n operation, EV charging facility operation and HVAC system operation of room j in building n with given Lagrangian multipliers λ can be defined as:

$$\begin{aligned} \min_{P_{k,n}^p, P_{k,n}^s} L_n^B, \text{ with } L_n^B &= \sum_{k=1}^K [(c_k^p \Delta k + \lambda_{k,n}) P_{k,n}^p \\ &\quad - (c_k^s \Delta k + \lambda_{k,n}) P_{k,n}^s + \lambda_{k,n} P_{k,n}^{PV} - \lambda_{k,n} P_{k,n}^e] \\ \text{s.t. } &(3) - (5). \end{aligned} \quad (18)$$

$$\begin{aligned} \min_{P_{k,n}^{EV}} L_n^V, \text{ with } L_n^V &= \sum_{k=1}^K -\lambda_{k,n} P_{k,n}^{EV} \\ \text{s.t. } &(6) - (11). \end{aligned} \quad (19)$$

$$\begin{aligned} \min_{a_{k,j}^R} L_j^R, \text{ with } L_j^R &= \sum_{k=1}^K -\lambda_{k,n} a_{k,j}^R, \forall j \in J_n^B \\ \text{s.t. } &(12) - (15). \end{aligned} \quad (20)$$

Therefore, the dual problem of **P1** is defined as:

$$\begin{aligned} \max_{\lambda} \Phi(\lambda), \text{ with } \Phi(\lambda) &= \sum_{n=1}^N \min_{P_{k,n}^p, P_{k,n}^s} L_n^B + \sum_{n=1}^N \min_{P_{k,n}^{EV}} L_n^V + \sum_{j=1}^J \min_{a_{k,j}^R} L_j^R. \end{aligned} \quad (21)$$

In fact, the sub-problems of EV charging facility operation for different buildings still have spatial coupled constraints caused by EVs trip between buildings. Thus, other Lagrangian multipliers μ for EV charging facility operation sub-problems are introduced to relax (6). The Lagrangian function of EV charging facility operation sub-problem in building n is defined as:

$$\begin{aligned} L_n^V &= \sum_{k=1}^K -\lambda_{k,n} P_{k,n}^{EV} \\ &\quad + \mu_{k,n} \left[P_{k,n}^{EV} - \sum_{i=1}^I L_{k,i,n} (a_{k,i}^{V,c} - a_{k,i}^{V,d}) P_i^V \right], \end{aligned} \quad (22)$$

where $\mu = \{\mu_{k,n}\}$, $\forall n \in N, \forall k \in K$ is the set of Lagrangian multipliers. Then the sub-problem of EV charging facility operation in building n can be rewritten as:

$$\begin{aligned} \min_{P_{k,n}^{EV}} L_n^{EVCF}, \text{ with } L_n^{EVCF} &= \sum_{k=1}^K (\mu_{k,n} - \lambda_{k,n}) P_{k,n}^{EV} \\ \text{s.t. } &(7). \end{aligned} \quad (23)$$

The sub-problem of EV i charging can be defined as:

$$\begin{aligned} \min_{a_{k,i}^{V,c}, a_{k,i}^{V,d}} L_i^{EV}, \text{ with } &L_i^{EV} = \sum_{k=1}^K \sum_{n=1}^N -\mu_{k,n} L_{k,i,n} (a_{k,i}^{V,c} - a_{k,i}^{V,d}) P_i^V \\ \text{s.t. } &(8) - (11). \end{aligned} \quad (24)$$

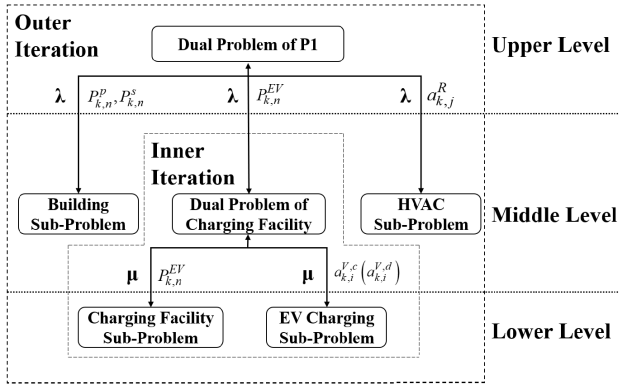


Fig. 2. The three-level optimization problem structure and information exchange mechanism.

Similarly, the dual problem of (19) is defined as:

$$\begin{aligned} & \max_{\mu} \Omega_n(\mu), \text{ with } \Omega_n(\mu) \\ & = \min_{P_{k,n}^{EV}} L_n^{EVCF} + \sum_{i=1}^I \min_{a_{k,i}^{V,c}, a_{k,i}^{V,d}} L_i^{EV}. \end{aligned} \quad (25)$$

After relaxing the constraints (2) and (6), the coordinated problem **P1** is converted into three levels as shown in Fig. 2. The dual problem of **P1** is in the upper level. In the middle level, there are three sub-problems which are the sub-problems of building operation, the sub-problems of HVAC system operation and the dual problems of EV charging facility operation sub-problem. The EV charging sub-problems are in the lower level. The sub-problems in middle and lower level do not have coupled constraints and can be solved independently.

The dual problem of **P1** can be rewritten as:

$$\begin{aligned} & \max_{\lambda} \Phi(\lambda), \text{ with } \Phi(\lambda) \\ & = \sum_{n=1}^N \min_{P_{k,n}^p, P_{k,n}^s} L_n^B + \sum_{n=1}^N \max_{\mu} \Omega_n(\mu) + \sum_{j=1}^J \min_{a_{k,j}^R} L_j^R. \end{aligned} \quad (26)$$

According to the duality theory [25], the optimal solution of the problem **P1** can be derived by solving the dual problem (26). Therefore, once the optimal Lagrangian multipliers are obtained, the central coordinator can broadcast them to HVAC systems and EVs. The HVAC systems and EVs can make the decision according to the optimal Lagrangian multipliers in a decomposition and coordination way.

Thus, a LR-based three-level structure is presented in Fig. 2. The coordinated problem can be solved under a double-layer iteration framework. The optimal Lagrangian multipliers are obtained by the sub-gradient method, which are theoretically guaranteed to converge to the optimal solution [26]. The information exchange mechanism between different levels is also shown in Fig. 2.

For the inner iteration, charging facility in buildings spread the Lagrangian multipliers μ to EVs for them to make decisions independently in the lower level. Then, μ are upgraded by solving the dual problems of EV charging facility operation sub-problem in the middle level according to the charging decision uploaded by EVs. The charging facility uploads the aggregated demand of EVs to central coordinator.

Algorithm 1 LR Framework-Based Iteration Algorithm

Step1. Initialization: Set $l=0, u=0$. Initialize the Lagrangian multipliers i.e., λ_0 and μ_0

Step2. Upgrade the solution of EVs: Solve EV charging sub-problem (24). Solve EV charging facility operation sub-problem (23).

Step3. Upgrade the Lagrangian multipliers μ_l : Obtain the gradient direction of μ_l by:

$$g_{k,n,l}^{\mu} = P_{k,n,l}^{EV} - \sum_{i=1}^I L_{k,i,n}(a_{k,i,l}^{V,c} - a_{k,i,l}^{V,d}) P_i^V.$$

Choose the step size s_l^{μ} which should satisfy:

$$0 < s_l^{\mu} < (L_n^{V,*} - L_n^{V,l}) / \|g_l^{\mu}\|^2,$$

where g_l^{μ} denotes the set of $g_{k,n,l}^{\mu}$, $L_n^{V,*}$ is the optimal value of Lagrangian function of EV charging facility operation sub-problem of building n .

Update the multipliers:

$$\mu_{k,n,l+1} = \mu_{k,n,l} + s_l^{\mu} g_{k,n,l}^{\mu}, l = l + 1.$$

If $\|\mu_{l+1} - \mu_l\| \leq \epsilon$, where ϵ is the accuracy requirement, go to Step4. Otherwise, go back to Step 2.

Step4. Construct a feasible solution: Construct the feasible solution for EV charging facility operation sub-problems.

Step5. Upgrade the solution of buildings: Solve HVAC system operation sub-problem (20).

Solve building operation sub-problem (18).

Step6. Upgrade the Lagrangian multipliers λ_u : Obtain the gradient direction of λ_u by:

$$\begin{aligned} g_{k,n,u}^{\lambda} = & P_{k,n,u}^p - P_{k,n,u}^s + P_{k,n}^{PV} \\ & - P_{k,n}^e - P_{k,n,l}^{EV} - \sum_{j \in J_n^B} a_{k,j,u}^R. \end{aligned}$$

Choose the step size s_u^{λ} which should satisfy:

$$0 < s_u^{\lambda} < (L^* - L_u) / \|g_u^{\lambda}\|^2,$$

where g_u^{λ} denotes the set of $g_{k,n,u}^{\lambda}$, L^* is the optimal value of Lagrangian function of system.

Update the multipliers:

$$\lambda_{k,n,u+1} = \lambda_{k,n,u} + s_u^{\lambda} g_{k,n,u}^{\lambda}, u = u + 1.$$

If $\|\lambda_u - \lambda_{u-1}\| \leq \epsilon$, where ϵ is the accuracy requirement, go to Step7. Otherwise, go to Step2.

Step7. Construct a feasible solution: Construct the feasible solution for **P1**.

For the outer iteration, the central coordinator spread the Lagrangian multipliers λ to the charging facility for inner iteration, buildings and HVAC systems for operation decision making in middle level. The λ are upgraded in the dual problem of **P1** in the upper level by central coordinator according to the demand uploaded by the buildings, charging facility and the decisions uploaded by HVAC systems. Therefore, the central coordinator can calculate the optimal multipliers according to limited information uploaded by buildings, HVAC systems and charging facility.

An LR framework-based iteration algorithm is developed. Note that in Algorithm 1, the central coordinator only needs to collect the aggregated demand of the buildings to upgrade the Lagrangian multipliers λ instead of all the global information. The occupants only need to upload their charging demand

to EV charging facility for upgrading Lagrangian multipliers μ instead of the trip plan and initial SoC. It can reduce the information transmission requirement and protect the privacy of occupants. Each EV and each HVAC system can make decision individually according to the Lagrangian multipliers after the central coordinator and EV charging facility upgrade them. Therefore, the computational time per iteration will not increase with the increased scale of EVs and HVAC systems. Although the computational complexity of the double-layer iterations does not depend on the scale of the problem. Its convergence efficiency and accuracy are hard to be guaranteed. Therefore, a DPLR algorithm is developed in next sub-section.

B. Dynamic Programming-Based LR Algorithm

Focused on the challenges mentioned above, a DPLR algorithm is developed in this sub-section. In this way, the dual problems of EV charging facility operation sub-problem (25), which are “max-min” structures problems, are transformed to “max” optimization problems of Lagrangian multipliers μ based on the optimal analytic expression of EV charging sub-problems. It can avoid the inner iteration in Algorithm 1 and improve the convergence efficiency. The details of DPLR algorithm are shown as below.

In the LR-based three-level structure developed in sub-section III-A, every single EV can obtain their optimal decision independently. But the dual problems of EV charging facility operation sub-problem have to be solved by the iteration of Lagrangian multipliers μ . The coordinated problem is solved under a double-layer iteration framework. The convergence efficiency is hard to be guaranteed. It is found that in the EV charging sub-problem (24), the decisions are made in discrete stages and the objective function is additive over time. Therefore, the optimal analytic expression of EV charging sub-problems can be derived by value iteration of dynamic programming [27]. The SoC and charging action of EVs can be defined as system state variables and decision variables, respectively. The cost function of EV i charging

sub-problem is derived as:

$$\begin{cases} J_{k,i}^{EV*}(S_{k,i}) \\ = \min_{a_{k,i}^{V,c}, a_{k,i}^{V,d}} \left\{ \sum_{n=1}^N -\mu_{k,n} L_{k,i,n} (a_{k,i}^{V,c} - a_{k,i}^{V,d}) P_i^V \right. \\ \left. + J_{k+1,i}^{EV*}(S_{k+1,i}(S_{k,i}, a_{k,i}^{V,c}, a_{k,i}^{V,d})) \right\}, \\ J_{K+1,i}^{EV*}(S_{K+1,i}) = 0, \end{cases} \quad (27)$$

where $S_{k+1,i}(S_{k,i}, a_{k,i}^{V,c}, a_{k,i}^{V,d})$ denotes the SoC of EV i at next time stage after execute $a_{k,i}^{V,c}$ and $a_{k,i}^{V,d}$ on $S_{k,i}$ at stage k . The cost function of the state at time stage k consists of cost-to-go and one-step cost.

Although the SoC of EV is a continuous variable, it can only transition to a few certain values with the given initial state, discrete energy consumption for trip, discrete charging and discharging actions. The feasible state of EV i 's SoC at time stage k with the given initial state and trip energy consumption for the cost function (27) can be denoted as (27a), shown at the bottom of the page. The k_i^a denotes the change of SoC of EV i caused by charging or discharging for a time step that $k_i^a = P_i^V \Delta k / E_i^{V, cap}$. The $k_{k,i}^c$ denotes the change of SoC of EV i caused by energy consumption of trip on the road for a time step at time stage k that $k_{k,i}^c = f_{k,i}(\Delta k) / E_i^{V, cap}$. As shown in Fig.3, the value in the green area is infeasible because of EV's battery capacity constraint. The states in yellow area is infeasible that is caused by EV charging and discharging power constraint. The other values in the white area are the values that can be transitioned to from the SoC of the EV with the certain initial state and trip energy consumption. Among them, the blue area denotes the EV is on the road which implies that the SoC of EV can only transition to the certain value based on the trip energy consumption.

Intuitively, at most three decisions can be made for a certain SoC at a certain time stage which are charging, discharging and idle. The objective is to choose the feasible decision that

$$S_{k,i} = \left\{ S_{K+1,i} + x k_i^a + \sum_{t=k}^K (1 - z_{t,i}^{EV}) k_{t,i}^c \mid \forall x \in \mathbb{Z}, |x| \leq K - k + 1, \underline{S} \leq S_{K+1,i} + x k_i^a + \sum_{t=k}^K (1 - z_{t,i}^{EV}) k_{t,i}^c \leq 1 \right\}. \quad (27a)$$

$$\begin{aligned} & J_{k,i}^{EV*}(S_{k,i,c}) \\ = & \begin{cases} J_{k+1,i}^{EV*}(S_{k+1,i,c+1}) - \sum_{n=1}^N \mu_{k,n} L_{k,i,n} P_i^V & \text{if } S_{k+1,i,c+1} \in S_{k+1,i}, z_{k,i}^{EV} \neq 0, \\ J_{k+1,i}^{EV*}(S_{k+1,i,c+1}) - \sum_{n=1}^N \mu_{k,n} L_{k,i,n} P_i^V \leq J_{k+1,i}^{EV*}(S_{k+1,i,c-1}) + \sum_{n=1}^N \mu_{k,n} L_{k,i,n} P_i^V, \\ J_{k+1,i}^{EV*}(S_{k+1,i,c+1}) - \sum_{n=1}^N \mu_{k,n} L_{k,i,n} P_i^V \leq J_{k+1,i}^{EV*}(S_{k+1,i,c}) \\ J_{k+1,i}^{EV*}(S_{k+1,i,c-1}) + \sum_{n=1}^N \mu_{k,n} L_{k,i,n} P_i^V \leq J_{k+1,i}^{EV*}(S_{k+1,i,c+1}) - \sum_{n=1}^N \mu_{k,n} L_{k,i,n} P_i^V, \\ J_{k+1,i}^{EV*}(S_{k+1,i,c-1}) + \sum_{n=1}^N \mu_{k,n} L_{k,i,n} P_i^V \leq J_{k+1,i}^{EV*}(S_{k+1,i,c}) \\ J_{k+1,i}^{EV*}(S_{k+1,i,c}) & \text{else.} \end{cases} \end{aligned} \quad (27b)$$

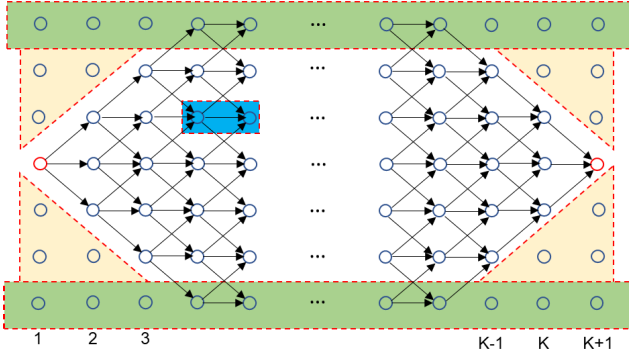


Fig. 3. State transition of EV SoC with given initial state and constraints.

minimizes the cost function among the three decisions. Thus, the cost function (27) can be transformed to (27b), as shown at the bottom of the previous page. The c th available discrete value of the SoC of EV i at time k is defined as $S_{k,i,c}$. Therefore, the neighboring states are denoted as $S_{k,i,c+1} = S_{k,i,c} + k_i^a$ and $S_{k,i,c-1} = S_{k,i,c} - k_i^a$. The optimal solution of single EV charging sub-problem (24) is equivalent with the cost of its initial state $J_{1,i}^{EV*}$ which can be derived independently according to the trip information and the initial state of SoC uploaded by occupant. The objective function of the dual problems of charging facility operation sub-problem (25) can be rewrite as (28). The first item is the optimal solution of the sub-problem of EV charging facility operation which should be on the bound of the charging facility capacity constraint that can be described as (29) - (31), where $z_{k,n}^{EVCF} \in \{0, 1\}$. The optimal analytic expression of EV charging sub-problem (27b) chooses the minimum value under three condition and it can be transformed to (32) - (36) as the constraints of the dual problems of charging facility operation sub-problem.

The problem (25) can thus be transformed to (28) - (36).

$$\Omega^*(\lambda) = \max_{\mu} \sum_{n=1}^N \sum_{k=1}^K (\mu_{k,n} - \lambda_{k,n}) P_{k,n}^{EV} + \sum_{i=1}^I J_{1,i}^{EV*} \quad (28)$$

$$s.t. \quad \mu_{k,n} - \lambda_{k,n} + (1 - z_{k,n}^{EVCF})M \geq 0, \quad (29)$$

$$\mu_{k,n} - \lambda_{k,n} - z_{k,n}^{EVCF}M < 0, \quad (30)$$

$$P_{k,n}^{EV} = (1 - 2 \times z_{k,n}^{EVCF}) P_n^{EV,cap}, \quad (31)$$

$$J_{k,i}^{EV*}(S_{k,i,c}) = +\infty, \text{ if } S_{k,i,c} \notin S_{k,i}, \quad (32)$$

$$J_{k,i}^{EV*}(S_{k,i,c}) = J_{k+1,i}^{EV*}(S_{k+1,i,c}), \quad (33)$$

$$\text{if } z_{k,i}^{EV} = 0, S_{k,i,c} \in S_{k,i}, \quad (34)$$

$$J_{k,i}^{EV*}(S_{k,i,c}) \leq J_{k+1,i}^{EV*}(S_{k+1,i,c}), \quad (35)$$

$$\text{if } S_{k,i,c} \in S_{k,i}, z_{k,i}^{EV} \neq 0, \quad (36)$$

$$J_{k,i}^{EV*}(S_{k,i,c}) \leq J_{k+1,i}^{EV*}(S_{k+1,i,c-1})$$

$$+ \sum_{n=1}^N \mu_{k,n} L_{k,i,n} P_i^V,$$

$$\text{if } S_{k,i,c} \in S_{k,i}, z_{k,i}^{EV} \neq 0, \quad (35)$$

$$J_{k,i}^{EV*}(S_{k,i,c}) \leq J_{k+1,i}^{EV*}(S_{k+1,i,c+1})$$

$$- \sum_{n=1}^N \mu_{k,n} L_{k,i,n} P_i^V,$$

$$\text{if } S_{k,i,c} \in S_{k,i}, z_{k,i}^{EV} \neq 0. \quad (36)$$

Algorithm 2 DPLR Algorithm

Step1. Initialization: Set $u=0$. Initialize the Lagrangian multipliers i.e., λ_0 .

Step2. Upgrade the solution of EVs:

Solve (28) s.t. (29)-(36).

Solve (24) with obtained μ .

Step3. Upgrade the solution of buildings : Solve HVAC system operation sub-problems (20).

Solve building operation sub-problems (18).

Step4. Check the convergence: Obtain the gradient direction of λ_u by:

$$g_{k,n,u}^{\lambda} = P_{k,n,u}^p - P_{k,n,u}^s + P_{k,n}^{PV} - P_{k,n}^e - P_{k,n,u}^{EV} - \sum_{j \in J_n^B} a_{k,j,u}^R.$$

Choose the step size s_u^{λ} which should satisfy:

$$0 < s_u^{\lambda} < (L^* - L_u) / \|g_u^{\lambda}\|^2,$$

where g_u^{λ} denotes the set of $g_{k,n,u}^{\lambda}$, L^* is the optimal value of Lagrangian function of system.

Update the multipliers:

$$\lambda_{k,n,u+1} = \lambda_{k,n,u} + s_u^{\lambda} g_{k,n,u}^{\lambda}, u = u + 1.$$

If $\|g_u^{\lambda} - g_{u-1}^{\lambda}\| \leq \epsilon$, where ϵ is the accuracy requirement, go to Step5. Otherwise, go to Step2.

Step5. Construct a feasible solution: Construct the feasible solution for **P1**.

The dual problems of charging facility operation sub-problem (25) with “max-min” structure are transformed to “max” problems (28) - (36) of Lagrangian multipliers μ with given λ . It can be solved directly to substitute the inner iteration in Algorithm 1 (Step 2-Step 4). The DPLR algorithm is developed to obtain the optimal solution of **P1**.

Therefore, the **P1** can be solved in a single-layer iteration structure shown as Algorithm 2 instead of the double-layer iteration structure in Algorithm 1 which can improve the convergence efficiency. The optimality is theoretically guaranteed by duality theory [25]. The optimal solution of the dual problem obtained from sub-gradient method may not satisfy all the constraints. So, a feasible solution has to be constructed to satisfy all the constraints. Actually, the duality gap always exists due to the discrete variables. However, the duality gap can be used to evaluate the performance of the feasible solution constructed. The optimal analytic expression of EV charging sub-problem can be derived independently. The computational complexity of solving the problem (25) do not depend on the scale of EVs consists in the system. Hence, the computational complexity is greatly reduced while dealing with the large-scale problem. In next section, the performance of coordinated optimization and the developed DPLR algorithm is evaluated by several case studies.

IV. CASE STUDIES AND NUMERICAL RESULTS

A. Description of Numerical Case Studies

In this section, the performance of the developed method is demonstrated using several numerical case studies with dif-

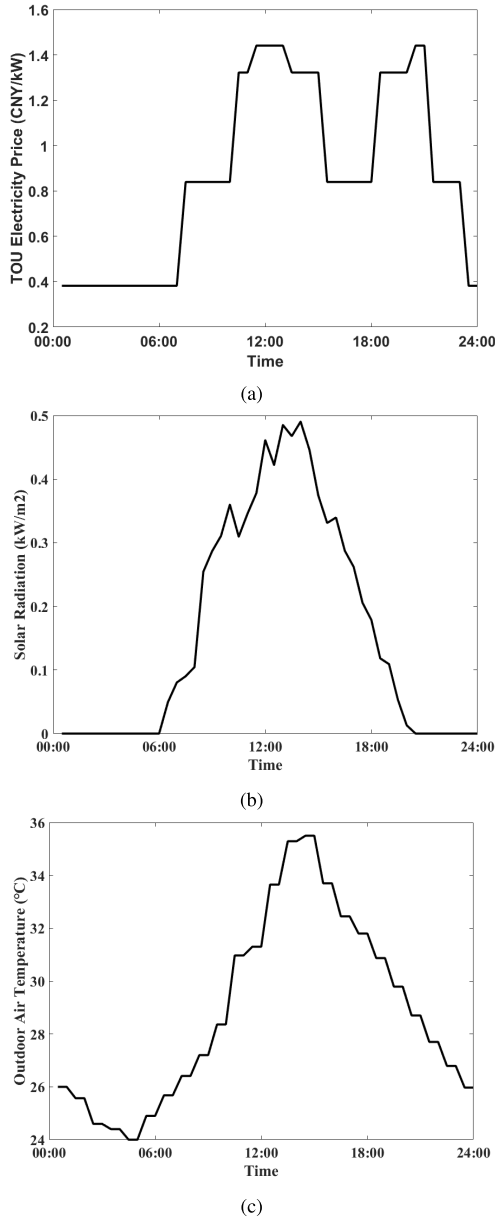


Fig. 4. (a) TOU tariffs, (b) Solar radiation, (c) Outdoor air temperature.

ferent scales. Since the performance of the developed method is generally related to the problem scale but not influenced by weather conditions, the trip rule of the occupants, and other system inputs, the coordinated optimization problem of 10 buildings with EVs in Beijing for a typical summer day is tested as an example. Fig. 4 shows the TOU electricity price of Beijing obtained from [28], the solar radiation and outdoor air temperature obtained from [29]. The photovoltaic generation output can be calculated based on solar radiation density and installed capacity [30]. The electricity selling price is set as $c_k^s = 0.3 \text{ CNY/kWh}$ which is according to the grid purchase price of Beijing [31], [32]. The scheduling horizon of the coordinated optimization problem is 24 hours and uniformly divided into 48 stages. The buildings are classified as 6 residential buildings, 3 office buildings and a commercial building. The area of photovoltaic generation installed on the roof-top of three types of buildings are 50 m^2 ,

TABLE I
EV NUMBER AND BUILDINGS PARAMETERS FOR 5 CASES

Case		1	2	3	4	5
EV Number		10	30	50	100	200
Area (m^2)	Residential	200	500	1000	2000	4000
	Office	200	200	500	1000	1000
	Commercial	600	600	1000	2000	2000
Room Number of Each Building	Residential	2	3	10	16	30
	Office	2	2	5	10	10
	Commercial	1	1	2	4	4
Total Power	Residential	10	15	50	80	150
Capacity of HVAC	Office	10	10	25	50	50
System (kW)	Commercial	20	20	50	100	100

50 m^2 and 100 m^2 , respectively. The parameters for the room indoor temperature dynamic model and the thermal comfort requirement of occupants are obtained according to [33]. EnergyPlus is used to simulate the demand profiles of other electrical equipments per unit area in buildings [34]. Assume that the rooms in residential buildings are occupied during 0:00-9:00 and 18:00-24:00, the rooms in office buildings and commercial building are occupied during 9:00-18:00 and 10:00-22:00, respectively.

The occupants are assumed evenly distributed live in residential building and work in office building. From 2015 Beijing transportation report [35], the time that people go to work and get off work follow normal distribution and can be described as $N(8:00, (1\text{h})^2)$ and $N(18:00, (1\text{h})^2)$. The trip time between different buildings is assumed to follow normal distribution $N(1\text{h}, (0.5\text{h})^2)$. Assume that occupants drive to office buildings from residential buildings in the morning and drive back to residential buildings in the end of the day of which half of them go to commercial building for shopping or entertainment and others go back home directly after work. The trip chain of EVs is simulated using Monte Carlo method according to the above patterns and assumed to be known in advance. The energy consumption rate, battery capacity and rated charging and discharging power of EVs are set as $f_{k,i}(\Delta t) = 5 \text{ kWh}$, $E_i^{V, \text{cap}} = 50 \text{ kWh}$ and $P_i^V = 20 \text{ kW}$.

The computational complexity of solving the coordinated optimization problem is mainly influenced by the scale of EVs and HVAC systems. As shown in TABLE I, 5 different scales of cases are designed to demonstrate the computational efficiency and accuracy of the developed DPLR algorithm. The scale of EVs and HVAC systems is increased from Case 1 to Case 5. Note that the number of occupants, each building areas, the capacity of HVAC systems are also increased along with the rise in the problem scale from Case 1 to Case 5. In following, the performance of the developed method is validated using Case 1 to Case 5 and also compared with that of the centralized algorithm and the traditional LR algorithm.

B. Performance Analysis of Coordinated Optimization Compared With Individual Optimization

In this subsection, the energy cost reduction of the coordinated optimization of the multi-building energy system with EVs is evaluated. The scheduling strategy obtained by the

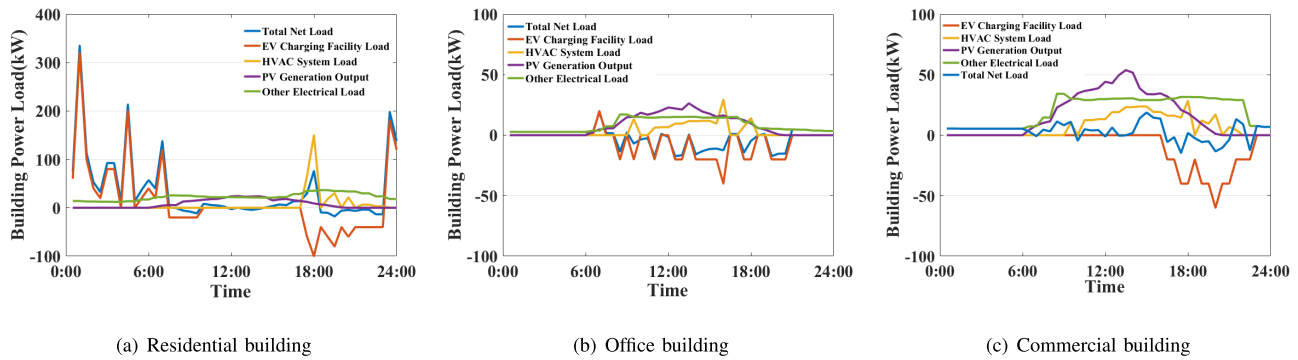


Fig. 5. Power loads of three different type buildings under coordinated optimization: (a) Residential building, (b) Office building, and (c) Commercial building.

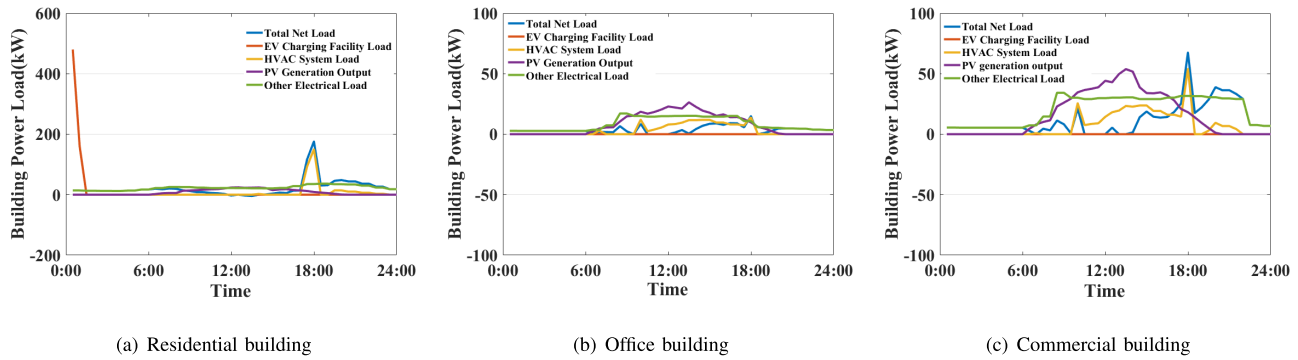


Fig. 6. Power loads of three different type buildings under individual optimization: (a) Residential building, (b) Office building, and (c) Commercial building.

TABLE II

TOTAL COST AND COST REDUCTION OF INDIVIDUAL OPTIMIZATION AND COORDINATED OPTIMIZATION

	Total Cost/CNY	Cost Reduction
Individual Optimization	5625.58	/
Coordinated Optimization	3133.98	44.30%

DPLR algorithm is compared with the strategy obtained by independent optimization of EVs and HVAC systems which is used as the benchmark. In independent optimization strategy, EVs optimal reduce their cost according to TOU electricity price. The HVAC systems optimal reduce building energy cost according to the load of other electrical equipments in buildings, PV supply and TOU electricity price.

TABLE II shows that the coordinated optimization of the multi-building energy system with EVs makes over 44.3% energy cost reduction compared with individual optimization. Fig. 5. shows the power load of a residential building, an office building and a commercial building respectively under the coordinated optimization strategy obtained by the DPLR algorithm while Fig. 6. shows them under independent optimization strategy. It can be found that under the strategy obtained from the DPLR algorithm, the EVs consume more electricity for charging in 0:00-8:00 and 23:00-24:00 with valley electricity price. And EVs feed electricity to buildings with peak electricity price (18:00-21:00). The independent optimization strategy never uses EVs as energy storage system. The coordination can greatly enhance the flexibility of the system. It makes a significant reduction in buildings' total load during the peak price period of the TOU tariff mechanism and

TABLE III

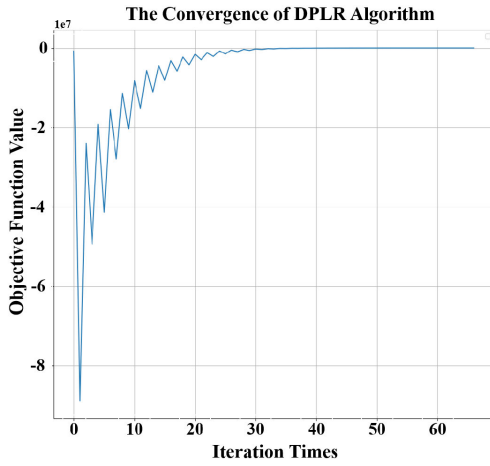
TOTAL COST AND COMPUTATIONAL TIME OF DIFFERENT ALGORITHM FOR 5 CASES

Case	Algorithm		
	Centralized	Traditional LR	DPLR
Total Cost (CNY)	1	-369.73	-355.95
	2	-108.11	-99.79
	3	410.63	440.26
	4	/	1593.56
	5	/	3156.40
Computational time (s)	1	54.90	977.08
	2	605.37	1263.82
	3	1270.03	1584.09
	4	/	1911.51
	5	/	5966.95

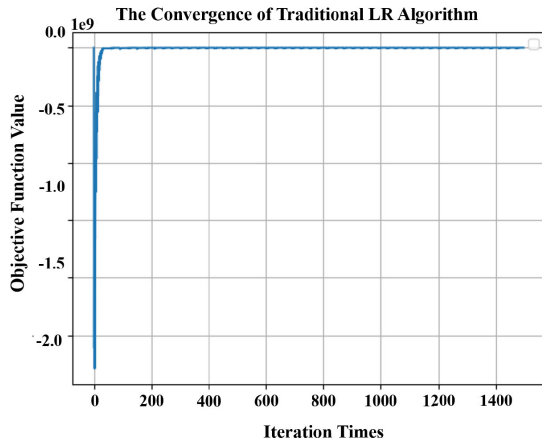
causes a reduction in the total cost of the multi-building energy system with EVs.

C. Performance Analysis of DPLR Algorithm

In this subsection, the performance of the developed DPLR algorithm is evaluated by comparing with the centralized algorithm and traditional LR algorithm based on the 5 cases in different scales mentioned above. The problems are solved by Python with Gurobi 9.0.1 on a Windows 10 PC which has a CPU with 2.6GHz main frequency and 8GB RAM. The total cost of the coordinated optimization problem and computational time are shown in TABLE III. The buildings considered in Case 1 to Case 5 are the same. Among all cases, EVs scales are different. Thus, the roof-top PV generation installed capacity is the same in all cases and the system can



(a) DPLR algorithm



(b) Traditional LR algorithm

Fig. 7. The convergence of DPLR algorithm and traditional LR algorithm for Case 5.

get income from selling the extra PV power back to the grid which is bigger than the energy cost in small cases. Therefore, the total costs of small cases (e.g., Case 1 and 2) are negative.

Compared with other algorithms, the results from TABLE III show that the developed DPLR algorithm has good scalability. Although the centralized algorithm has better computational efficiency in small cases, it cannot deal with large-scale problems because of the curse of dimensionality. Due to the two-level iteration, the traditional LR algorithm takes more time and iterations to solve the same problem and has worse convergence performance. The DPLR algorithm is three times faster without sacrificing too much accuracy compared with the centralized algorithm in medium cases. In the large-scale cases that cannot be solved by the centralized algorithm, the DPLR algorithm can get better result with 52% computational time saving compared with the traditional LR algorithm.

Besides, the accuracy of the DPLR algorithm is also demonstrated. The results obtained by the DPLR algorithm are only 2%-4% worse than the optimal result which is obtained by the centralized algorithm while the results of the traditional LR algorithm are 3%-8% worse. The duality gaps of DPLR algorithm and traditional LR algorithm are shown in TABLE IV. It is found that the duality gaps of them are

TABLE IV
THE CONVERGENCE OF DPLR ALGORITHM AND
TRADITIONAL LR ALGORITHM

Case	Algorithm	
	Traditional LR	DPLR
1	10.98%	8.21%
2	9.01%	7.07%
3	8.37%	7.40%
4	5.93%	5.58%
5	5.66%	4.37%

4%-8% and 5%-11%, respectively. The convergence processes of these two algorithms for Case 5 (as an example) are shown in Fig. 7. The results show that the convergence efficiency of DPLR is over 20 times better than traditional LR algorithm, the similar results are also found in other cases.

V. DISCUSSION

A. Comparison of DPLR Algorithm With Traditional LR Algorithm and Centralized Algorithm

On solving an optimization problem with N buildings, J rooms and I EVs by centralized algorithm, the state space at time stage k is $S_k \subset \mathbb{R}^I \times \mathbb{R}^J$. The decision space of EVs is $2^{I \times K} \times 2^{I \times K}$ and the decision space of HVAC systems is $A^R = (A_1^R, A_2^R, \dots, A_K^R) \subset \mathbb{R}^{J \times K}$. The decision space in the centralized problem is the decision space of EVs times the decision space of HVAC systems that will exponentially grow with the increasing scale of the problem. In DPLR algorithm, the HVAC system operation sub-problems and EV charging facility operation sub-problems in middle level are decoupled by relaxation of (2) and their decision space is thus decoupled. Moreover, the EV charging sub-problems in lower level are decoupled by relaxation of (6) that lead to the decision space of them is reduced from $2^{I \times K} \times 2^{I \times K}$ to $I(2^K \times 2^K)$ which will linearly grow with the increasing scale of the problem. Therefore, the computational complexity of DPLR is greatly reduced compared with centralized algorithm. The optimality of DPLR algorithm is theoretically guaranteed by duality theory. Although the non-convexity caused by the binary variables consist in the model may lead to duality gap, the case studies show that it only sacrifices little accuracy while getting the solution three times faster.

Compared with traditional LR algorithm, the DPLR algorithm can solve the optimization problem in a single-layer iteration structure which can reduce the computational complexity and improve convergence efficiency. The optimal analytic expression of EV charging sub-problems respect to the Lagrangian multipliers are derived independently. Thus, the dual problems of EV charging facility sub-problem can be solved directly and the computational complexity of solving the problem do not depend on the scale of EVs and it also avoids the redundant iteration. The numerical results show that the DPLR algorithm can get better solution with less computational time.

B. Occupant's Information Privacy Protection

In the centralized algorithm, the central coordinator has to collect all the global information (including the trip plans and initial state of charge of EVs, and the power supplied by roof-top photovoltaic generation) from the occupants to make optimal decision and scheduling the demand of EVs and HVAC systems to satisfy occupant's requirement and reduce the energy cost under TOU tariffs.

In practice, occupants may not be willing to share all the information to central coordinator due to privacy concern. Thus, the DPLR algorithm is developed in this paper. In this way, the occupants of EVs and HVAC systems can make their own decision in a decomposition and coordination way according to the Lagrangian multipliers. Instead of uploading the private information such as trip plan and get the optimal action signal from the centralized way, based on DPLR algorithm, the central coordinator only needs to collect the information of the aggregated demand of EVs and HVAC systems uploaded by occupants to calculate the optimal Lagrangian multipliers. The trip plan and demand information of single occupant is not necessary for coordinately scheduling. It can partly protect the occupant's privacy.

C. Information Exchange Mechanism

In DPLR algorithm, the central coordinator broadcasts the Lagrangian multipliers λ to EV charging facility and rooms in buildings. EV charging facility spreads the Lagrangian multipliers μ to EVs. The EVs upload their demand information to EV charging facility and buildings collect and upload the aggregated demand information of EV charging facility and rooms to central coordinator to calculate the optimal Lagrangian multipliers μ and λ . By introducing the Lagrangian multipliers μ and λ , the sub-problems of HVAC systems and the sub-problems of EVs are decoupled due to the relaxation of the power balance constraints. In this way, EVs and HVAC systems can make their own decision individually in a decomposition and coordination way according to the Lagrangian multipliers λ and μ which are treated as price signal.

VI. CONCLUSION

A coordinated optimization problem of the multi-building energy system with EVs is investigated in this paper. The problem is formulated as a centralized optimization problem to capture the spatio-temporal coupling and interaction between EVs and HVAC systems caused by occupant's thermal comfort requirement and mobility. The problem is converted into a three-level optimization problem base on LR framework in a decomposition and coordination way to overcome the computational complexity and global information requirement challenge. However, the convergence and accuracy of solving the optimization problem with the three-level structure is hard to be guaranteed due to the double-layer iterations. A DPLR algorithm is developed to transform the dual problems of EV charging facility operation sub-problem from "max-min" structure problem to "max" optimization problems in order to solve it directly. The numerical result shows that the

coordination of EVs and building HVAC system can reduce 44% of the energy cost by utilizing their load flexibility. The developed DPLR algorithm is at least three times faster than the centralized algorithm with less than 4% worse accuracy. It also can save 52% with better accuracy compared with the traditional LR algorithm.

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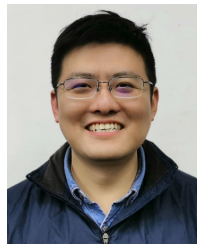
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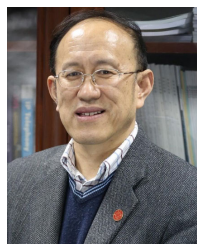


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